Jack Birkin, PhD

Data Scientist

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Profile

Data scientist utilizing linear modeling and machine learning techniques to carry out cutting-edge astronomy research. Effectively worked as an independent researcher and as part of large international collaborations. Over 10 years of experience programming with Python and Jupyter notebooks. Highly skilled in problem solving, data visualization, independent work, collaborative work and presenting.

Skills

- 10+ years of Python programming experience and data visualization with Jupyter Notebooks
- Data cleaning and preprocessing, feature engineering, linear modeling
- Database analysis, including selection and cross-matching (pandas, SQL, astropy)
- Statistical analysis, model evaluating, hypothesis testing
- Project leadership, academic paper writing, presenting and communicating technical concepts
- Web development: HTML, CSS, AWS S3/EC2, Linux

Work Experience

Postdoctoral Research Associate
Texas A&M University, College Station, TX | Aug 2022 – Present

- Developed and applied linear models to analyze high-dimensional datasets and infer underlying system properties from spectral data using Python and Markov Chain Monte Carlo methods.
- Used SQL to perform cross-matching of large datasets, integrating multi-source observations to support high-confidence candidate identification.
- Developed a Python-based pipeline to aggregate and analyze data from multiple sources, improving weak signal detection by a factor of 3.9 through stacking 15 independent datasets, confirming the absence of large-scale motion-driven effects.
- Developed and published an open-source Python module to model and enhance data visualization of 3D structured datasets, improving model selection and validation. Utilized Git for version control and collaborative development.
- Authored 3 peer-reviewed papers and co-authored 10 others within 3 international collaborations, totaling over 180 citations.

Graduate Researcher
Durham University, UK | Oct 2018 – Jul 2022

- Led the planning, data processing and analysis of a 200-hour large program, including the development of a sophisticated Python pipeline for efficient workflow execution.
- Developed models for analyzing emission data from 50 high-dimensional datasets, achieving 7.7% precision in gas content measurements. Created and implemented a method to identify and eliminate spurious signals in complex 3D data.
- Analyzed large-scale simulations with over a billion data points and implemented inverse modeling techniques to recover signals from initial conditions.

Education

PhD, Physics | Durham University | Oct 2018 – Aug 2022
MPhys, Physics & Astronomy (First Class Honours) | Durham University | Oct 2014 – Jun 2018

Certifications

AWS Certified Cloud Practitioner | Amazon Web Services
AWS Certified Solutions Architect | Amazon Web Services
Machine Learning Specialization | Stanford University & DeepLearning.AI